

Beyond Demographics: Enhancing Cultural Value Survey Simulation with Multi-Stage Personality-Driven Cognitive Reasoning

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Why Survey Simulation?

Traditional human surveys are essential for social science research, but face significant limitations:

- **High costs:** \$1M+ for large-scale studies (World Values Survey)
- **Time-consuming:** Years to collect & analyze responses
- **Limited scalability:** Difficult to test hypotheses across cultures

AI-based simulation offers a cost-effective alternative for:

- Rapid hypothesis testing
- Cross-cultural value analysis
- Exploratory research at scale

The Challenge

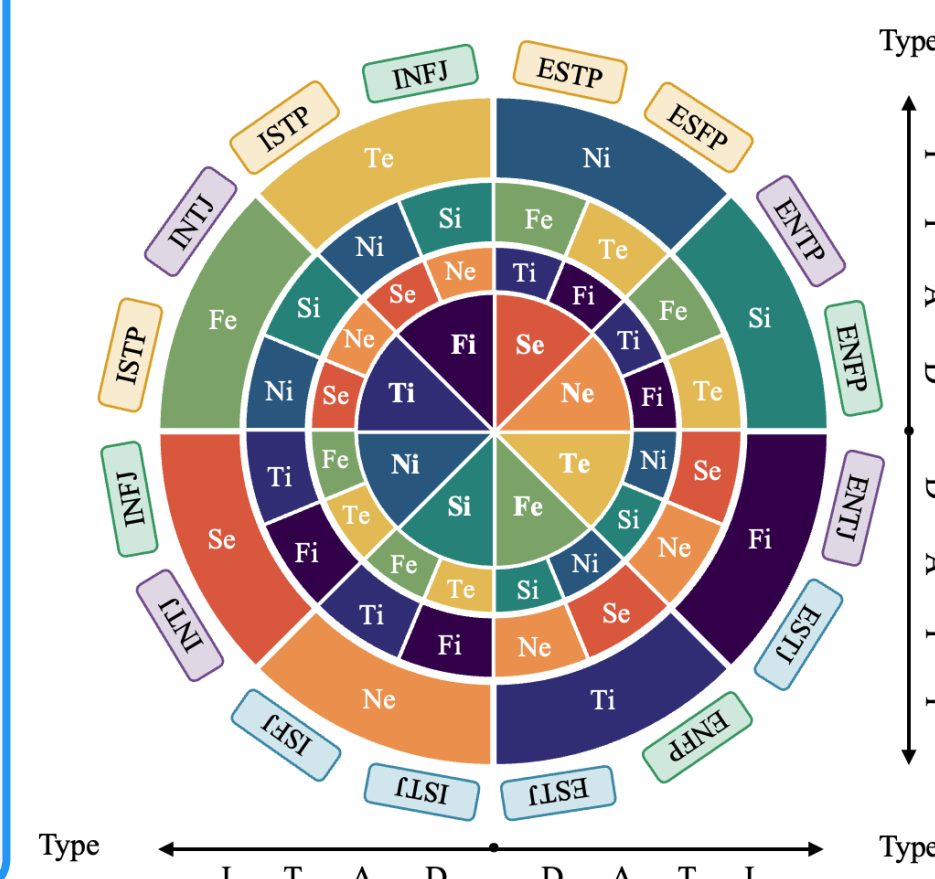
However, current simulation methods rely on demographics alone:

- **Limited accuracy** (30-36%)
- **No psychological modeling**
- **Poor interpretability**

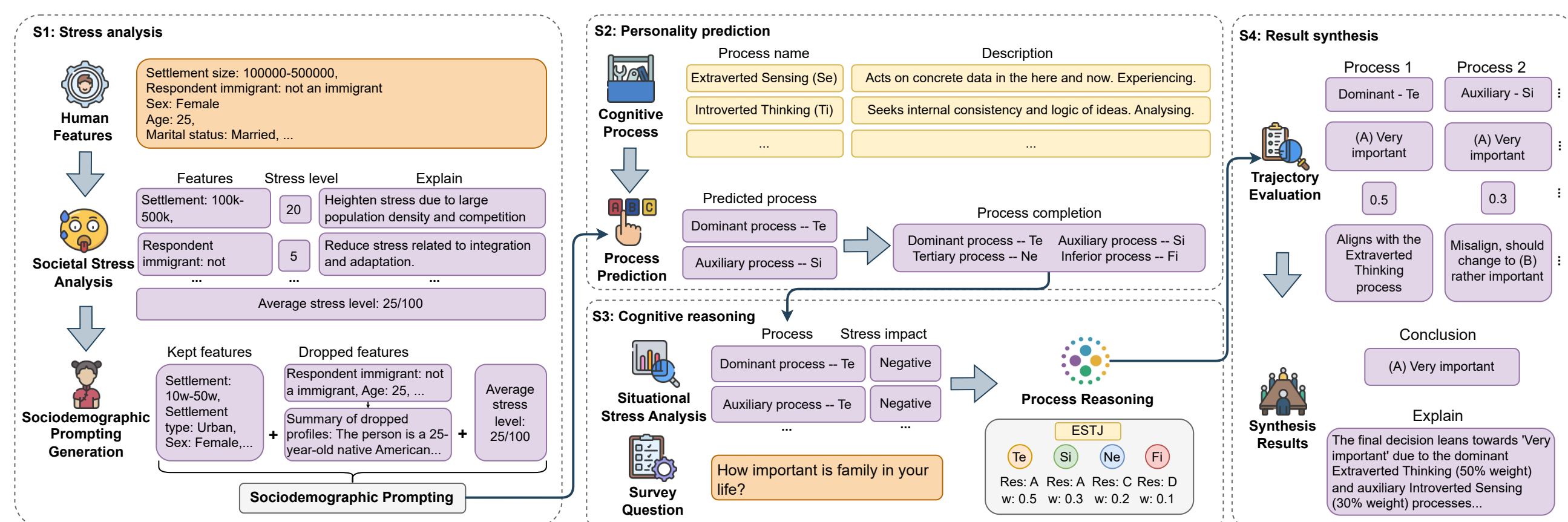
Our Insight: Psychology Meets AI

Core Insight (Left Figure: Personality Compass):

- Human survey responses are influenced by **mental status** (stress levels)
- Different personality types have **distinct cognitive processing patterns**
- **MBTI Type Dynamics** provides hierarchical framework



MARK Pipeline



1. Stress Analysis

Evaluate life-situational pressures (0-100 scale)
Filter high-stress & personality-relevant features

2. Personality Prediction

Infer MBTI functions from demographic context
Predict Dominant & Auxiliary → derive Tertiary & Inferior

3. Cognitive Reasoning

Multi-process weighted simulation
Model normal vs. overused functioning patterns

4. Result Synthesis

Evidence-based integration with weights:

$$w_D > w_A > w_T > w_I$$

Key Innovation:

We model *how* personality types process information, not just *what* demographics predict

Experimental Settings

Dataset

- World Values Survey (Wave 7)
- 30 demographic features
- 20 clusters (K-means)

Models

- GLM-4-air
- GPT-4o
- Doubao-1.5-pro

Baselines

- Demo. only
- Demo. + Ideology (+ Opinion)
- Zhao et al. (2024)

Metrics

- ACC: Response accuracy
- Distribution Div (1-JSD_↑, EMD_↓)
- Cohen's κ : Group agreement

Results: +10% Accuracy

Method	GLM-4-air				GPT-4o				Doubao-1.5-pro			
	ACC	1-JSD	EMD	κ	ACC	1-JSD	EMD	κ	ACC	1-JSD	EMD	κ
Demo.+Ideo.	25.49	0.354	0.074	0.02	32.30	0.408	0.076	0.09	30.75	0.456	0.069	0.02
+Opinion	25.06	0.431	0.067	0.11	33.07	0.407	0.074	0.09	31.45	0.472	0.070	0.08
Zhao et al.	26.98	0.381	0.045	0.05	36.96	0.465	0.061	0.12	24.23	0.358	0.036	0.05
MARK	33.69	0.435	0.089	0.15	38.11	0.488	0.083	0.15	46.98	0.520	0.056	0.09
Improve.	+6.7	+0.004	—	+0.04	+1.2	+0.023	—	+0.03	+15.5	+0.048	—	+0.01

Table 1: Comprehensive results on U.S. World Values Survey (sampled distribution, $p < 0.05$). Green highlights show MARK's improvements over the best baselines.

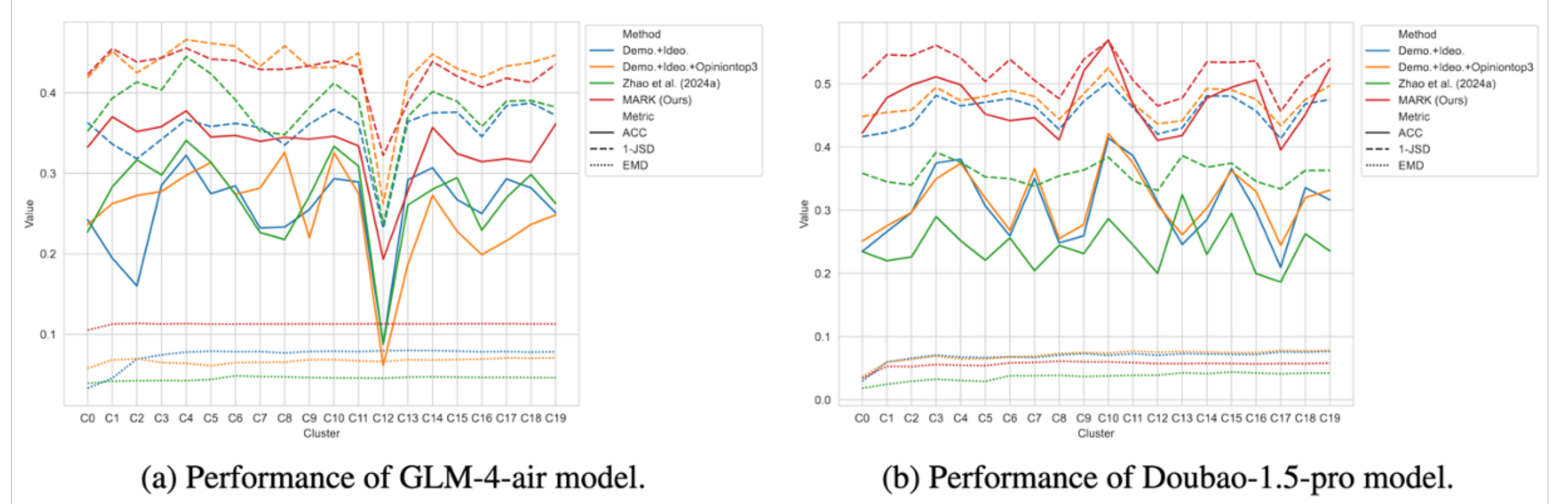


Figure 1: Group-level performance across 20 demographic clusters

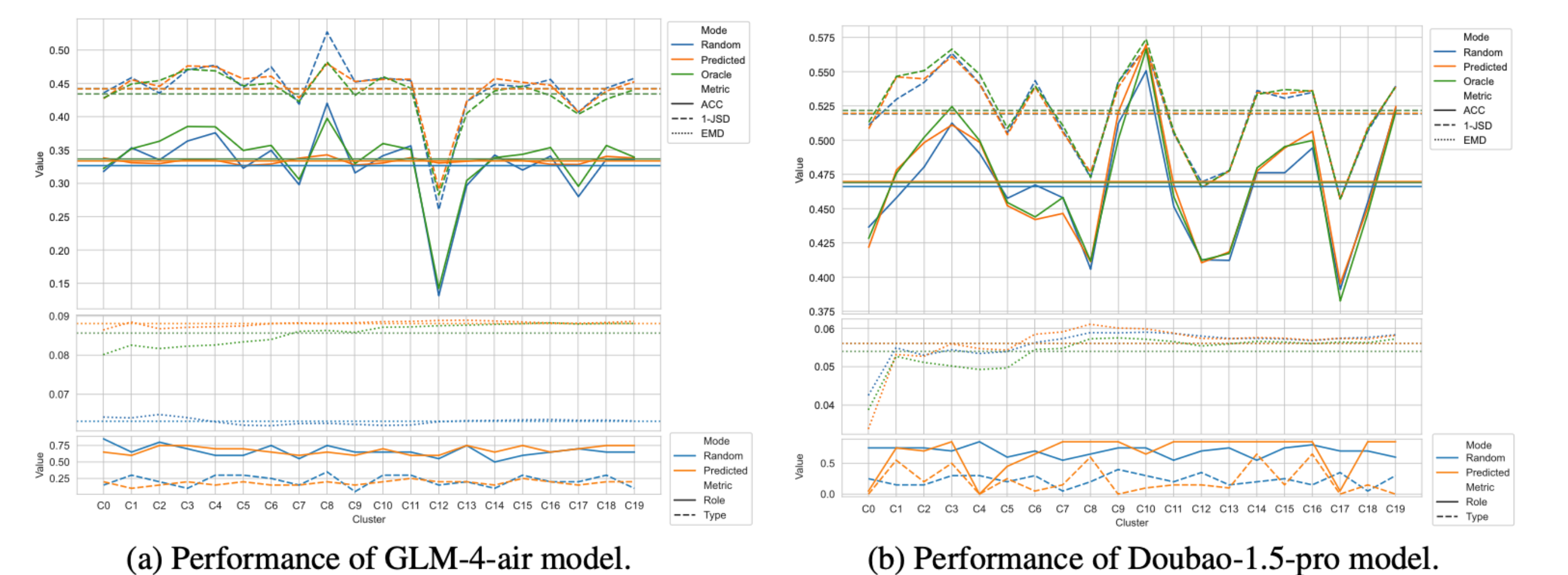


Figure 2: Stable performance across personality assignment strategies (Predicted, Random, Oracle)

Key Findings

- **Consistent improvement** across all models
- **Better distribution alignment** (highest 1-JSD: 0.520)
- **3-7× higher agreement** (Cohen's κ : 0.15 vs. 0.02)
- **Robust to personality uncertainty**

Cross-Cultural Success

MARK generalizes well on Chinese Culture.

Method	Chinese Prompts			English Prompts		
	ACC	1-JSD	EMD	ACC	1-JSD	EMD
<i>Doubao-1.5-pro on Chinese Culture</i>						
Demo.+Ideo.	30.35	0.5007	0.0810	26.98	0.5039	0.0888
+Opinion	30.55	0.5128	0.0830	27.44	0.5244	0.0884
Zhao et al.	30.28	0.4407	0.0897	32.24	0.4515	0.0923
MARK	45.16	0.6010	0.0861	44.08	0.6001	0.0839

Table 2: Cross-cultural and cross-lingual validation

Key Contributions

- First MBTI-based multi-stage reasoning for survey simulation
- 10% accuracy improvement with enhanced interpretability
- Robust to personality uncertainty

- Successful cross-cultural generalization (U.S.-China)
- Opens new directions for psychology-informed AI
- Cost-effective alternative to traditional human surveys